A One-Vertex-Deletion Conjecture for Square-Sum Hamiltonian Cycles

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Abstract

Let G_n join distinct integers in $[n]$ when their sum is a square. We conjecture that $G_n - d$ has a Hamiltonian cycle for every $n \geq 71$ and every vertex d, strengthening known ordinary square-sum cycles. We prove that 71 is the first possible threshold, establish minimum-degree and edge-count asymptotics, and give an exact one-rotation closure criterion. A specified 71-cycle yields a local counterexample to one-rotation deletion repair, while longer rotations succeed. Independently checked witnesses cover all 4775 deletion instances with $71 \leq n \leq 120$. These results separate arithmetic sparsity, local repair obstructions, and finite existence evidence from the still-open infinite assertion.

Preprint; not externally peer reviewed. AI use and verification are described at the end.

Introduction and research strategy

The recorded starting point was the familiar square-sum circle problem, followed by a literature check establishing that its ordinary all-large-orders version was already solved. The exploratory question was then changed in one precise way: can the arithmetic adjacency rule tolerate one arbitrarily missing label? This is a robustness question about the same graph family, not a new name for ordinary Hamiltonicity.

Small searches suggested the number 71, and a local degree obstruction explained why it could not be smaller. This interaction matters: a threshold fitted to successful examples alone could be accidental, whereas vertex 50 at order 70 gives a necessary obstruction independently of any solver. Later searches accumulated deletion witnesses and attempted block constructions; those later achievements should not be attributed to the initial observation.

Strategy and limitations. We establish the sharp necessary degree threshold, count the edges, and locate the graph in a sparse regime with degrees of order $\sqrt{n}$. We then examine exactly what a simple deletion repair would require. An explicit 71-cycle supplies a hand-checkable failure of repair by a single rotation, even though a longer rotation sequence succeeds. This distinction separates a false local shortcut from the surviving global conjecture. The finite certificate covers every deletion through order 120; proving an arithmetic extension invariant or a uniform rotation theorem remains the infinite task.

1 The fault-tolerant assertion

Throughout, graphs are finite and simple, and a cycle has at least three vertices. Put $[n] = \{1, \ldots, n\}$ and

$$V(G_n) = [n], \qquad uv \in E(G_n) \iff u \neq v \text{ and } u + v \text{ is a perfect square.}$$

The labels are retained after deletion. In particular, the problem on $[n] \setminus \{d\}$ is not generally the same as the problem on $[n-1]$.

Conjecture 1.1. For every $n \geq 71$ and every $d \in [n]$, the induced graph $G_n - d$ has a Hamiltonian cycle.

Each deletion is allowed its own ordering. Equivalently, the $n-1$ remaining numbers can be arranged around a circle with all adjacent sums, including the closing sum, perfect squares. This does not require that one fixed cycle survive every deletion.

2 What is already known

Ordinary square-sum necklaces are established objects. The appendix to Berlekamp–Guy [1] credits Robert Gerbicz's January 2018 construction with proving existence for every $n \geq 32$; it also records square-sum chains for every $n \geq 25$. We use that statement as background and cite the appendix rather than presenting the ordinary necklace theorem as a new conjecture. In particular, deleting $d = n$ is already settled whenever $n \geq 33$.

Conjecture 1.1 asks for robustness under *every* vertex deletion, a stronger condition than ordinary Hamiltonicity. Focused searches through 14 November 2026, using square-sum necklace, vertex deletion, fault Hamiltonicity, and the threshold 71, did not locate its exact statement in the checked sources. This does not establish priority. Graph-labeling papers using the phrase "square-sum graph" can concern a different sum-of-squared-labels condition and should not be treated as results on G_n without checking their definitions.

3 The sharp necessary degree threshold

Lemma 3.1. For $v \in [n]$,

$$\deg_{G_n}(v) = \lfloor \sqrt{v+n} \rfloor - \lfloor \sqrt{v} \rfloor - \mathbf{1}_{\{2v \text{ is a square}\}}.$$

Proof. Potential neighbors have the form $s^2 - v$ with $v < s^2 \leq v + n$. The last term removes the possible self-neighbor $s^2 = 2v$. □

Proposition 3.2. For every $n < 71$, there is a deletion d for which $G_n - d$ is not Hamiltonian. For every $n \geq 71$, $\delta(G_n) \geq 3$.

Proof. For $n \leq 3$ the remaining graph has fewer than three vertices. For $4 \leq n \leq 70$, the following vertex has degree at most two, as follows directly from Lemma 3.1:

n	4–14	15–22	23–45	46–48	49	50–70
v	1	2	18	32	49	50

If this vertex has degree two, delete a neighbor; if it has degree at most one, delete any other vertex. Its degree in the remaining graph is at most one, excluding a Hamiltonian cycle.

The finite evaluation of the same formula for $1 \leq v \leq 71$ gives $\delta(G_{71}) = 3$; all 71 values are checked in the supplied program. Existing vertex degrees never decrease when a new vertex is added. For the newly added vertex n, let $k = \lfloor \sqrt{n} \rfloor$. For $n \geq 64$, $k \geq 8$ and

$$\deg_{G_n}(n) = \lfloor \sqrt{2n-1} \rfloor - k \geq 3,$$

because $2k^2 - 1 \geq (k+3)^2$. Induction from 71 completes the proof. □

At order 70, vertex 50 has just the neighbors 14 and 31; the apparent sum $50 + 50 = 100$ is not an edge. At order 71, vertex 71 supplies its third neighbor. Thus 71 is forced by an explicit local obstruction. Proposition 3.2 proves necessity of the proposed threshold, not sufficiency: minimum degree three alone does not imply Hamiltonicity.

4 Rotation witnesses with a small certificate

For a path $(v_1, \dots, v_L)$, if $v_L v_i$ is an edge with $i < L$, reversing its terminal segment gives

$$(v_1, \dots, v_i, v_L, v_{L-1}, \dots, v_{i+1}).$$

This is again a spanning path on the same vertex set. This elementary rotation operation underlies Hamiltonian-path methods associated with Pósa [2]; the present search uses it as an operation whose validity is directly checked, not as a general existence theorem.

The program `verify_square_cycles.py` starts with the following 32-cycle, also displayed in Berlekamp–Guy [1, Figure 2]:

$$(4, 21, 28, 8, 1, 15, 10, 26, 23, 2, 14, 22, 27, 9, 16, 20,$$
$$29, 7, 18, 31, 5, 11, 25, 24, 12, 13, 3, 6, 30, 19, 17, 32).$$

It constructs ordinary cycles up to 120 by attaching the next vertex and applying breadth-first rotations at either endpoint. Deleting d from a cycle leaves a spanning path in $G_n - d$; the same rotation search attempts to close it. The implementation imposes a limit of 20000 visited states per closure. All instances in the claimed range succeeded; the largest observed search used 765 states. This observed maximum is not a uniform theorem about the algorithm.

Proposition 4.1 (Finite certificate)**.** For every $71 \le n \le 120$ and every $d \in [n]$, $G_n - d$ has a Hamiltonian cycle.

Certificate and checking rule. The file `square_deletion_cycles_71_120.json.gz` stores a cycle for each of the $\sum_{n=71}^{120} n = 4775$ pairs (n, d). The independent final check verifies its length, distinctness, exact vertex set $[n] \setminus \{d\}$, and every consecutive and closing sum using integer square roots. The generation script reruns the construction and these checks using only the Python standard library. Its summary is `square_verification.json`. Thus successful search is converted to explicit existence witnesses; a failed or interrupted search would not constitute a negative result. □

The compressed witness file is about 66 kB. The claims require neither factorial storage nor a GPU. Earlier project prose reports a wider range and infinite constructions, but the present finite proposition is deliberately restricted to the freshly supplied certificate.

5 How sparse are the square-sum graphs?

The degree obstruction is local, but the graph is not of bounded degree as n grows. The following estimates locate the difficulty more precisely.

Proposition 5.1. The exact number of edges is

$$|E(G_n)| = \sum_{s=2}^{\lfloor\sqrt{2n}\rfloor} \begin{cases} \lfloor (s^2-1)/2 \rfloor, & s^2 \le n+1, \\ \lfloor (2n+1-s^2)/2 \rfloor, & s^2 > n+1. \end{cases}$$

Consequently,

$$|E(G_n)| = \frac{2}{3}(\sqrt{2}-1)n^{3/2} + O(n), \qquad \delta(G_n) = (\sqrt{2}-1)\sqrt{n} + O(1).$$

Every vertex has degree less than $\sqrt{n+1}$.

Proof. For a fixed square s^2, count the pairs $u < v$ with $u + v = s^2$. If $s^2 \le n + 1$, choose $1 \le u \le \lfloor (s^2 - 1)/2 \rfloor$. Otherwise the lower bound is $u \ge s^2 - n$, giving the second expression. No edge is counted twice.

Replacing floors by real quantities contributes $O(\sqrt{n})$. Summing squares, or comparing to the corresponding integrals, gives

$$\frac{1}{2}\int_0^{\sqrt{n}} t^2\,dt + \frac{1}{2}\int_{\sqrt{n}}^{\sqrt{2n}} (2n - t^2)\,dt + O(n) = \frac{2}{3}(\sqrt{2} - 1)n^{3/2} + O(n).$$

For degrees, Lemma 3.1 implies

$$\deg(v) \ge \sqrt{v + n} - \sqrt{v} - 2 \ge (\sqrt{2} - 1)\sqrt{n} - 2.$$

On the other hand, $\delta(G_n) \le \deg(n) = \lfloor \sqrt{2n - 1} \rfloor - \lfloor \sqrt{n} \rfloor$, giving the matching upper estimate. Finally

$$\deg(v) < \sqrt{v + n} - \sqrt{v} + 1 \le \sqrt{n + 1},$$

since the difference of square roots decreases with v. □

For example, the exact edge counts at $n = 71, 120, 1000$ are $159, 359, 8711$. The average degree is asymptotic to $\frac{4}{3}(\sqrt{2} - 1)\sqrt{n}$. Thus growing minimum degree coexists with vanishing edge density. For $n \ge 5$, even the maximum degree is below $n/2$, so the familiar sufficient conditions of minimum degree at least $n/2$, or nonadjacent degree sums at least n, cannot establish Hamiltonicity here. Their failure does not imply a negative answer; it specifies why dense-graph criteria do not settle this arithmetic family.

6 Exactly what a one-rotation repair can do

Lemma 6.1 (One-rotation closure)**.** Let $P = (v_1, \ldots, v_L)$ be a spanning path. If its endpoints are nonadjacent, it can be closed after one nontrivial endpoint rotation if and only if, for some $2 \le i \le L - 2$, both $v_L v_i$ and $v_1 v_{i+1}$ are edges.

Proof. The two edges produce the cycle

$$(v_1, \ldots, v_i, v_L, v_{L-1}, \ldots, v_{i+1}, v_1).$$

Conversely, a rotation from v_L breaks exactly $v_i v_{i+1}$, with new free endpoint v_{i+1}, so closing requires exactly the second edge. A rotation from the other endpoint gives the same test after reversing the path. Trivial rotations cannot change nonadjacent endpoints into adjacent ones. □

Proposition 6.2. At $n = 71$, deleting vertex 50 from any Hamiltonian cycle cannot be repaired merely by joining its former two neighbors. Furthermore, some deletion paths cannot be closed with only one endpoint rotation.

Proof. The neighbors of 50 in G_{71} are $14, 31, 71$, with pairwise sums $45, 85, 102$, none square. Therefore the two neighbors used by any cycle cannot be joined after removing 50.

For the second assertion use the following explicit cycle:

$$\begin{aligned}(&26, 38, 43, 57, 7, 2, 62, 59, 22, 27, 37, 12, 69, 52,\\ &48, 1, 24, 25, 11, 5, 31, 50, 14, 67, 33, 16, 65, 35,\\ &29, 71, 10, 54, 46, 18, 63, 58, 42, 39, 61, 20, 44, 56,\\ &8, 28, 36, 64, 17, 32, 49, 51, 70, 30, 34, 47, 53, 68,\\ &13, 3, 6, 19, 45, 4, 60, 21, 15, 66, 55, 9, 40, 41, 23).\end{aligned}$$

Its consecutive and closing sums are squares and it uses [71] exactly once. Delete 1 and orient the resulting path from 24 to 48. These endpoints have sum 72. The available neighbors of the last endpoint 48 are $16, 33, 52$; their successors along this path are respectively $65, 16, 48$. None belongs to the available neighbor set $\{12, 25, 40, 57\}$ of the first endpoint 24. Lemma 6.1 excludes every one-rotation closure. □

For this particular cycle, a complete breadth-first exploration until the first closure for each deleted label yields:

Minimum rotations	0	1	2	3	4
Deletion positions	3	7	18	23	20

In particular the displayed deletion of 1 needs four rotations in this operation model. The program `verify_extensions_v2.py` retains the initial path and a repaired cycle, and its breadth-first ordering checks the absence of a shorter repair. This is a statement about one specified starting cycle and allowed operations, not a lower bound over every possible initial cycle. It disproves the simple repair rule while supporting the need for richer boundary configurations. Together with the 72-insertion obstruction below, it explains why successful finite rotation searches have not yet supplied an induction theorem.

7 Why the infinite step remains difficult

A naive insertion induction fails already when adding 72. Its neighbors in G_{71} are $9, 28, 49$, whose pairwise sums $37, 58, 77$ are not squares. Therefore no edge of any old cycle can simply be replaced by a two-edge path through 72. A proof must permit more substantial rerouting.

One possible strategy is to maintain a finite collection of path configurations with specified boundary labels and prove closure under an arithmetic enlargement operation. Such a theorem must cover every deletion position, including deletions inside the interfaces, and must supply base cases and a proof that every sufficiently large order is reached. Alternatively, one could prove enough expansion to guarantee that endpoint rotations close. At present there is no uniform bound on the reachable endpoint set or on the search depth. The sparse arithmetic graph also has many structured nonedges, so a degree calculation cannot replace that missing argument. The value of the conjecture is precisely the proposed passage from the known ordinary cycles to an arithmetic family with one-vertex fault tolerance.

Provenance and use of AI

This is a research draft, not a peer-reviewed publication. The problem arose in an AI-assisted exploratory project. AI tools were used substantially for literature searches, mathematical derivations, verification programs, and drafting. The proofs and bounded checks are supplied for independent scrutiny; neither an exhaustive novelty claim nor a general proof beyond the stated results is asserted.

References

[1] E. Berlekamp and R. K. Guy. Fibonacci Plays Billiards. arXiv:2002.03705 (2020), especially Figure 2 and the appendix on square-sum necklaces. https://arxiv.org/abs/2002.03705.

[2] L. Pósa. A theorem concerning Hamilton lines. *Magyar Tud. Akad. Mat. Kutató Int. Közl.* 7 (1962), 225–226. https://real.mtak.hu/201261/.